Why Hamiltonian Mechanics?

Predicting the next instant from position and momentum

Discovering Hamilton's equations before deriving them

How to read this tutorial

Pause at each boxed result and ask what it says geometrically or physically. The derivation is part of the explanation, not a hurdle to skip.

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1 The question that motivates the formalism

Lagrangian mechanics describes a configuration using q and a velocity $\dot{q}$. It also introduces the conjugate momentum

$$p = \frac{\partial L}{\partial \dot{q}}.$$

For an ordinary particle, $p = m\dot{q}$, so position and momentum contain the same instantaneous information as position and velocity.

This suggests a practical question:

What we actually need

If the particle is currently at (q, p), how do we find its position and momentum a moment later?

For a short interval Δt,

$$q(t + \Delta t) \approx q(t) + \dot{q}\,\Delta t, \qquad p(t + \Delta t) \approx p(t) + \dot{p}\,\Delta t.$$

We therefore need two rates:

$$\dot{q} = f(q, p, t), \qquad \dot{p} = g(q, p, t).$$

No partial derivatives of the Hamiltonian have been assumed. We have merely identified the information required to evolve the state.

2 Discovering the equations for a familiar particle

For a particle in a potential $V(q)$, define

$$H(q, p) = \frac{p^2}{2m} + V(q).$$

We already know

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = F = -\frac{\mathrm{d}V}{\mathrm{d}q}.$$

Now differentiate H:

$$\frac{\partial H}{\partial p} = \frac{p}{m} = \dot{q}, \qquad \frac{\partial H}{\partial q} = \frac{\mathrm{d}V}{\mathrm{d}q} = -\dot{p}.$$

Hamilton's equations, discovered

$$\boxed{\dot{q} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial q}.}$$

The two rates needed to predict the next instant are already contained in the two partial derivatives of a single function.

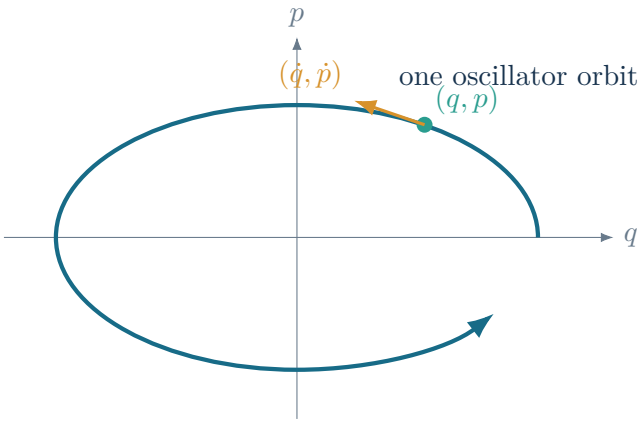
3 Why this is significant

Newton's equation is usually second order:

$$m\ddot{q} = F(q).$$

Hamilton's formulation replaces it with two coupled first-order equations for the state (q, p). This is not less information: specifying q and p is equivalent to specifying q and $\dot{q}$ for a regular mechanical system.

The pair (q, p) is a point in *phase space*. As time changes, the state traces a curve there.



At every point, Hamilton's equations assign the tangent vector

$$(\dot{q}, \dot{p}) = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q} \right).$$

The Hamiltonian is therefore a rule for the flow through phase space.

4 Why change variables from velocity to momentum?

The Lagrangian is naturally a function of $(q, \dot{q}, t)$. We want a new function of (q, p, t). Merely replacing symbols is not enough because p is defined by a derivative of L.

The required change is the Legendre transform:

$$\boxed{H(q, p, t) = p\dot{q} - L(q, \dot{q}, t),}$$

where $\dot{q}$ is re-expressed in terms of (q, p, t) using

$$p = \frac{\partial L}{\partial \dot{q}}.$$

What the Legendre transform does

It replaces the slope-like variable $\dot{q}$ by the conjugate slope p. The correction term $p\dot{q}$ preserves the information while changing which variable is independent.

For

$$L = \frac{1}{2} m\dot{q}^2 - V(q), \qquad p = m\dot{q},$$

we have $\dot{q} = p/m$ and

$$\begin{aligned} H &= p\frac{p}{m} - \left[\frac{1}{2} m \left(\frac{p}{m} \right)^2 - V(q) \right] \\ &= \frac{p^2}{2m} + V(q) = T + V. \end{aligned}$$

This familiar equality relies on ordinary quadratic kinetic energy and a velocity-independent potential; H is not automatically "the energy" in every model.

5 General derivation of Hamilton's equations

Begin with

$$H = p\dot{q} - L.$$

Take a total differential:

$$\mathrm{d}H = \dot{q}\ \mathrm{d}p + p\ \mathrm{d}\dot{q} - \mathrm{d}L.$$

Since

$$\mathrm{d}L = \frac{\partial L}{\partial q}\,\mathrm{d}q + \frac{\partial L}{\partial \dot{q}}\,\mathrm{d}\dot{q} + \frac{\partial L}{\partial t}\,\mathrm{d}t$$

and $p = \partial L/\partial \dot{q}$, the two terms containing $\mathrm{d}\dot{q}$ cancel:

$$\mathrm{d}H = \dot{q}\ \mathrm{d}p - \frac{\partial L}{\partial q}\,\mathrm{d}q - \frac{\partial L}{\partial t}\,\mathrm{d}t.$$

The Euler–Lagrange equation gives

$$\frac{\partial L}{\partial q} = \dot{p}.$$

Therefore

$$\mathrm{d}H = \dot{q}\ \mathrm{d}p - \dot{p}\ \mathrm{d}q - \frac{\partial L}{\partial t}\,\mathrm{d}t.$$

But $H = H(q, p, t)$ also implies

$$\mathrm{d}H = \frac{\partial H}{\partial q}\,\mathrm{d}q + \frac{\partial H}{\partial p}\,\mathrm{d}p + \frac{\partial H}{\partial t}\,\mathrm{d}t.$$

Comparing independent coefficients yields

$$\boxed{\dot{q} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial q}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}.}$$

6 Many degrees of freedom

For coordinates q_i and momenta

$$p_i = \frac{\partial L}{\partial \dot{q}_i},$$

define

$$H(\boldsymbol{q}, \boldsymbol{p}, t) = \sum_i p_i \dot{q}_i - L.$$

Then

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.}$$

The phase space has dimension $2n$: one coordinate and one conjugate momentum for each degree of freedom.

7 Worked example: harmonic oscillator

For

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}kq^2,$$

Hamilton's equations give

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = -kq.$$

Differentiate the first equation and insert the second:

$$\ddot{q} = \frac{\dot{p}}{m} = -\frac{k}{m}q.$$

Thus

$$\ddot{q} + \omega^2 q = 0, \qquad \omega = \sqrt{k/m}.$$

The constant-energy curves satisfy

$$\frac{p^2}{2m} + \frac{1}{2}kq^2 = E,$$

which are ellipses in the (q,p) plane. The oscillator moves around one ellipse without crossing to another energy.

8 Conservation of the Hamiltonian

Along a trajectory,

$$\begin{aligned}\frac{\mathrm{d}H}{\mathrm{d}t} &= \frac{\partial H}{\partial q}\dot{q} + \frac{\partial H}{\partial p}\dot{p} + \frac{\partial H}{\partial t} \\ &= (-\dot{p})\dot{q} + (\dot{q})\dot{p} + \frac{\partial H}{\partial t} \\ &= \frac{\partial H}{\partial t}.\end{aligned}$$

Therefore, if H has no explicit time dependence,

$$\boxed{H = \text{constant along the motion.}}$$

9 Poisson brackets

For functions $A(q,p,t)$ and $B(q,p,t)$ define

$$\{A,B\} = \frac{\partial A}{\partial q}\frac{\partial B}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial B}{\partial q}.$$

Hamilton's equations can be written

$$\dot{q} = \{q,H\}, \qquad \dot{p} = \{p,H\}.$$

More generally,

$$\boxed{\frac{\mathrm{d}A}{\mathrm{d}t} = \{A,H\} + \frac{\partial A}{\partial t}.}$$

If A has no explicit time dependence and $\{A,H\} = 0$, then A is conserved.

The basic bracket is

$$\{q,p\} = 1.$$

This compact algebra becomes the classical precursor of commutators in quantum mechanics.

10 Phase-space geometry

In two-dimensional phase space the velocity field is

$$\boldsymbol{v}_{\mathrm{ph}} = \left(\frac{\partial H}{\partial p}, -\frac{\partial H}{\partial q}\right).$$

Its divergence is

$$\frac{\partial}{\partial q}\left(\frac{\partial H}{\partial p}\right) + \frac{\partial}{\partial p}\left(-\frac{\partial H}{\partial q}\right) = 0$$

when mixed derivatives commute. Small phase-space areas are neither compressed nor expanded by Hamiltonian flow. This is the local heart of Liouville's theorem.

11 When the Legendre transform can fail

To solve $p = \partial L/\partial \dot{q}$ for $\dot{q}$, the velocity Hessian must be invertible. In one dimension this means

$$\frac{\partial^2 L}{\partial \dot{q}^2} \neq 0.$$

If it is singular, the system contains constraints and needs a more careful Hamiltonian treatment. The standard equations remain correct for regular Lagrangians.

12 A numerical viewpoint

Hamilton's equations are an initial-value system. Given (q_0, p_0), a simple first step is

$$q_{n+1} = q_n + \Delta t\, \frac{\partial H}{\partial p}(q_n, p_n), \qquad p_{n+1} = p_n - \Delta t\, \frac{\partial H}{\partial q}(q_n, p_n).$$

This explicit Euler rule is easy but does not preserve Hamiltonian geometry well over long times. Symplectic methods alter the update order to respect phase-space structure more faithfully.

13 What to remember

1. Position and momentum specify the instantaneous state.
2. Predicting the next instant requires $\dot{q}$ and $\dot{p}$.
3. For a familiar particle, these rates are the two partial derivatives of H with opposite signs.
4. The Legendre transform proves that the same structure follows generally from a regular Lagrangian.
5. The Hamiltonian generates a flow through phase space.

The central insight

Hamilton's equations are not arbitrary formulas involving partial derivatives. They say that a single scalar function contains the two local rates needed to move the complete state (q, p) forward in time.